LESSON 6.3

Explore Numerical Expressions with Exponents

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write and evaluate numbers in exponent form using a base and a power.

WHY THIS MATTERS You're a sound engineer at a music studio. Each time you flip a switch, the volume doubles. The starting volume is 1 unit. After flipping the switch 3 times, the volume is $2^3 = 2 \times 2 \times 2 = 8$ units — that's 8 times louder! How loud does it get after more flips?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Exponent *Spanish: Exponente*	A small number that tells how many times to multiply a number by itself.	In 9^4, the exponent is 4.
Base *Spanish: Base*	The number that gets multiplied by itself.	In 9^4, the base is 9.
Power *Spanish: Potencia*	A number written with a base and an exponent, like 2^3.	9^4
Evaluate *Spanish: Evaluar*	To find the value by putting numbers in for the letters.	Evaluate 3^4: write 3 × 3 × 3 × 3, then multiply step by step: 9 × 9 = 81
Repeated multiplication *Spanish: Multiplicación repetida*	Multiplying the same factor again and again.	9 × 9 × 9 × 9
Expanded form *Spanish: Forma desarrollada*	A power written out as a product of its factors.	2^3 = 2 × 2 × 2

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Write 9 × 9 × 9 × 9 as a power.

1. The repeated factor is 9, so the base is 9.
2. Count the 9s: there are four, so the exponent is 4.
3. Check: the exponent counts factors, not × signs (there are only 3 × signs).

Answer: 9^4; base 9, exponent 4

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now let's find 5^2. How many times do we multiply 5 by itself?

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- In this power, the base is _____ and the exponent is _____ .
- The exponent tells me to use the base as a factor _____ times.
- I read this power as _____ to the _____ power.
- A power shows repeated _____ , not repeated _____ .

Word bank power · base · exponent · factor · repeated multiplication · expanded form · squared · cubed · evaluate · product · read · times

WATCH OUT!

A common mistake

A common mistake in Powers and Exponents is multiplying the base by the exponent instead of using the base as a repeated factor

LESSON 6.3

Explore Numerical Expressions with Exponents

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 Circle the letter of the best answer. Show how you know.

Which expression shows 5^3 as repeated multiplication?

(A) $3 \times 3 \times 3 \times 3 \times 3$

(B) $5 \times 5 \times 5$

(C) 5×3

(D) $5 + 5 + 5$

HINT In 5^3 the base is 5 and the exponent is 3. Ask: which number repeats, and how many times?

SHOW YOUR WORK

2 Complete the table. Show how you found each value.

Write each power as repeated multiplication, then find the value.

Power	Repeated Multiplication	Value
3^2	3×3	
2^4	$2 \times 2 \times 2 \times 2$	

HINT The middle column already shows each power as repeated multiplication — use it!

SHOW YOUR WORK

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

What is the value of 4^3?

(A) 7

(B) 12

(C) 43

(D) 64

HINT Look at the small raised 3. That exponent counts how many 4's get multiplied — it is not a number to multiply 4 by.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

What is the value of 6^2?

(A) 8

(B) 12

(C) 36

(D) 62

HINT The little 2 is the exponent. It counts how many 6's you multiply — it's not a number to add or multiply by.

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Evaluate: $3 \times (4 + 2)^2$

(A) 36 (B) 54

(C) 108 (D) 144

HINT There are three jobs here: parentheses, an exponent, and multiplication. Order of operations decides which comes first.

Rewrite the whole expression each line	What I simplified

1. **Grouping** *Inside ( ) first.*
2. **Exponents** *Then powers.*
3. **× ÷** *Left to right.*
4. **+ −** *Left to right.*

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Expression** 5^3
2. **Student's thinking** 5×3
3. **Calculate** $= 15$

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Does the exponent tell you to multiply the base BY the exponent, or to use the base as a factor that many times?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How does the value change when you increase the exponent by 1? Why?

SENTENCE STARTER When the exponent increases by 1, the value is multiplied by ___ because the base is used as a factor ___ more time.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 6.3

Explore Numerical Expressions with Exponents

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Evaluate each exponential expression. Show the repeated multiplication and the final value.

Power	Repeated Multiplication	Value
4^3	$4 \times 4 \times 4$	
2^6	$2 \times 2 \times 2 \times 2 \times 2 \times 2$	
5^3	$5 \times 5 \times 5$	
10^4	$10 \times 10 \times 10 \times 10$	

SHOW YOUR WORK

2 **Write the name of the correct group on each line.**

Sort each power by whether its value is greater than 50 or not.

Groups: Value > 50 | Value ≤ 50

_____ 2^5

_____ 3^4

_____ 7^2

_____ 10^2

_____ 5^3

_____ 4^2

3 **Write the letter of the matching item on each line.**

Match each power to its value.

_____ 2^5	(A) 32
_____ 3^3	(B) 27
_____ 4^3	(C) 64
_____ 10^3	(D) 1000
_____ 5^2	(E) 25
_____ 2^7	(F) 128

4 **Decide whether each pair balances. Check Yes or No, then explain.**

Compare: 3^4 vs. 4^3

Left side	Right side	Balanced?

EXPLAIN YOUR CHOICE

PART 2 Explain and correct

reasoning

5 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Expression** 2^4
2. **Student's thinking** 2×4
3. **Calculate** $= 8$

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How does the value change when you increase the exponent by 1? Why?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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LESSON 6.3

Explore Numerical Expressions with Exponents

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Compare the values when the base and exponent are swapped. What do you notice?

Expression A	Value A	Expression B	Value B	Which is greater?
2^5		5^2		
2^4		4^2		
3^4		4^3		

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the Exponent Error

1. **Expression** 2^3
2. **Student's thinking** 2×3
3. **Calculate** $= 6$

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

A new song from the studio is shared online, and the number of shares doubles every hour. Starting with 1 share, write a power expression for the number of shares after 6 hours. Explain why exponential growth is so much faster than adding the same number each hour.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

How does the value change when you increase the exponent by 1? Why?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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